

LEARNING INTENTION

I can establish and use the formula for the area of a rectangle.

SUCCESS CRITERIA

- ☐ I know the area formula: length $\times$ width
- ☐ I can calculate the area of rectangles
- ☐ I can solve real-world area problems

STRATEGY BOX

Area of a rectangle = **length $\times$ width**. Always check your units — if length and width are in metres, your area is in square metres (m^2).

WORKED EXAMPLE

A rectangle is 8m long and 5m wide.

$$\text{Area} = 8 \times 5 = \mathbf{40 \text{ m}^2}$$

PART A — WARM UP

- Length 6cm, width 4cm. Area = _____
- Length 10m, width 3m. Area = _____
- Length 9cm, width 9cm (a square). Area = _____
- If area = 24cm^2 and length = 6cm, what is the width?

PART B — CORE SKILLS

- A rectangle has length 12.5m and width 6m. Area = _____
- A room is 4.2m by 3.5m. What is its floor area?
- A rectangular garden has an area of 36m^2 and a length of 9m. Find the width.
- Show your working for Question 7 below.

AREA HELPER

Area is always measured in **square units** (cm^2 , m^2 , km^2), unlike perimeter which is measured in plain units (cm, m). If you know the area and one side, divide by that side to find the missing one.

WORKING SPACE

PART C — CHALLENGE

9. Real-World Context: A farmer wants to plant grass on a rectangular paddock 50m long and 30m wide. Each bag of grass seed covers 100m^2 . How many bags does the farmer need?

10. Reasoning: Two rectangles both have an area of 24m^2 , but different length and width. Give two different possible sets of dimensions, and explain why they can have the same area.

REFLECTION

Today I learned...

I CAN CHECKLIST

- ☐ I know the area formula for rectangles
- ☐ I can find a missing side from area
- ☐ I can solve real-world area problems

TEACHER FEEDBACK

Strength: _____

Next Step: _____

PARENT TIP

Measure a room or table at home and calculate its area together.

TEACHER TIP

Use grid paper so students can see the formula matches counting unit squares.

ANSWER KEY (FOR PARENT / TEACHER USE)

Q1: $6 \times 4 = 24\text{cm}^2$

Q2: $10 \times 3 = 30\text{m}^2$

Q3: $9 \times 9 = 81\text{cm}^2$

Q4: $24 \div 6 = 4\text{cm}$

Q5: $12.5 \times 6 = 75\text{m}^2$

Q6: $4.2 \times 3.5 = 14.7\text{m}^2$

Q7: $36 \div 9 = 4\text{m}$

Q8: Working shown as above

Q9: $50 \times 30 = 1500\text{m}^2$; $1500 \div 100 = 15$ bags

Q10: e.g. $6\text{m} \times 4\text{m}$ and $8\text{m} \times 3\text{m}$ both = 24m^2 — area depends on the product, not a fixed shape